Did Canada's 2024 Tariff on Chinese Steel Change Where Canada Buys Its Steel?

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Summary

In October 2024, Canada added a 25% tax on steel imported from China. This report asks whether that tax changed where Canada buys its steel.

What happened. China's share of Canada's steel imports fell from about 6% before the tax to under 4% afterwards. The drop also shows up when imports are measured by weight instead of value. The decline remains when comparing the same winter months one year apart, making a purely seasonal explanation less likely. Imports from China themselves fell by about 39% compared with the same winter a year earlier, so this was not just other countries selling more.

Was it the tax? The tax applied only to some steel products. China's share fell more for the taxed products than for the untaxed ones: about 2 percentage points more, or roughly a third of China's earlier share. Before the tax, the two groups of products moved in step, which makes this comparison more convincing.

How sure can we be? Not very. Only four months passed between the tax and the start of a 2025 trade dispute between the United States and Canada, which disrupted steel trade again. With so little clean data, the estimate is not precise. Depending on choices in the calculation, it lands just on one side or the other of the usual line for statistical significance, and in a comparison that keeps the 167 products separate its likely range includes zero. Several checks point in the same direction, but none settles the question on its own.

Who gained? Comparing taxed and untaxed products for every supplier, Japan is the most likely gainer: its share of taxed products rose more than its share of untaxed products, and this holds up after allowing for testing several countries at once, though not once its earlier growth is also allowed for. For the United States and the many smaller suppliers, the data are too noisy to tell.

What it does not show. A smaller Chinese share does not by itself mean the tax was good for Canada.

1. The question

Canada buys steel from abroad when it is cheaper than making it at home. A tax on imports from one country makes that country's steel more expensive, so buyers may switch to suppliers elsewhere. Economists call this switching "trade diversion"; it follows from the offshoring model taught in international trade courses.

On August 26, 2024, Canada announced a 25% tax on steel and aluminum from China, and the tax started on October 22, 2024. This report asks three questions:

1. Did China's share of Canada's steel imports fall after the tax?
2. If so, which countries gained?
3. Was the change caused by the tax, or by other trends and policies?

Several other steel policies followed soon after, which makes it harder to isolate the effect of the tax:

Date	What happened
Aug 26, 2024	Canada announces a 25% tax on Chinese steel and aluminum
Oct 22, 2024	The tax starts
Mar 12-13, 2025	The U.S. puts a 25% tariff on Canadian steel; Canada responds with a 25% tariff on U.S. steel
Jun 27, 2025	Canada limits how much steel can come in at normal tariff rates from countries without a trade agreement with Canada, including China and Taiwan (Japan, South Korea and Vietnam are not affected)
Jul 31, 2025	A 25% tax on goods containing steel first made in China
Aug 1, 2025	The import limits are tightened and extended to most trade-agreement partners other than the U.S. and Mexico
Dec 26, 2025	The import limits are changed again
Jun 28, 2026	The import limits are changed again

2. Data

I used Statistics Canada's monthly records of Canada's steel imports from January 2023 to August 2026. The records cover 167 types of steel products and show both the value (in Canadian dollars) and the weight of each import. I downloaded the totals for imports from all countries, plus separate figures for six major suppliers: China, the United States, South Korea, Japan, Vietnam and Taiwan.

The code checks the files as it reads them. It stops with an error if a month is missing, if a value is missing or negative, if a unit other than kilograms or tonnes appears, or if the same product appears twice in the same month.

3. Measuring each country's share

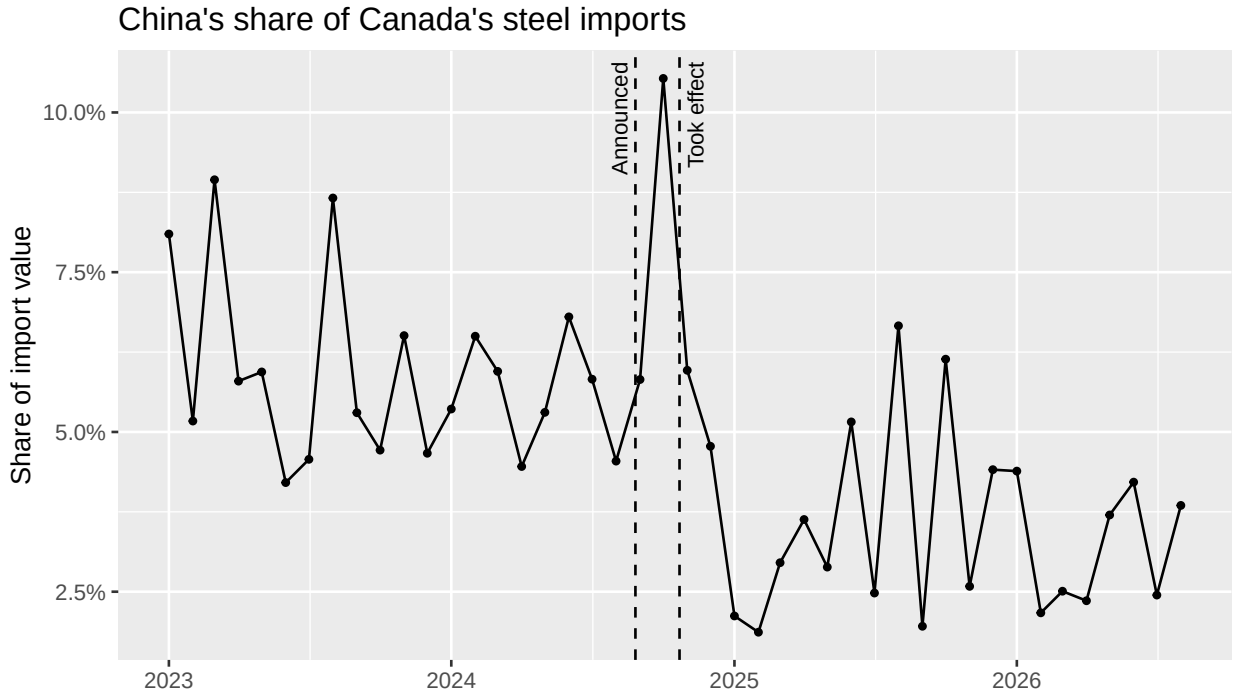
Each country's share is the value of steel Canada bought from it in a month, divided by the value of all the steel Canada imported that month. I also calculate the same share by weight.

As a check, the six suppliers together account for between 60% and 82% of Canada's steel imports in every month, so the figures add up sensibly.

Table 2: Share of Canada's monthly steel imports supplied by the six countries together

Minimum	Average	Maximum
60.4%	68.8%	82.0%

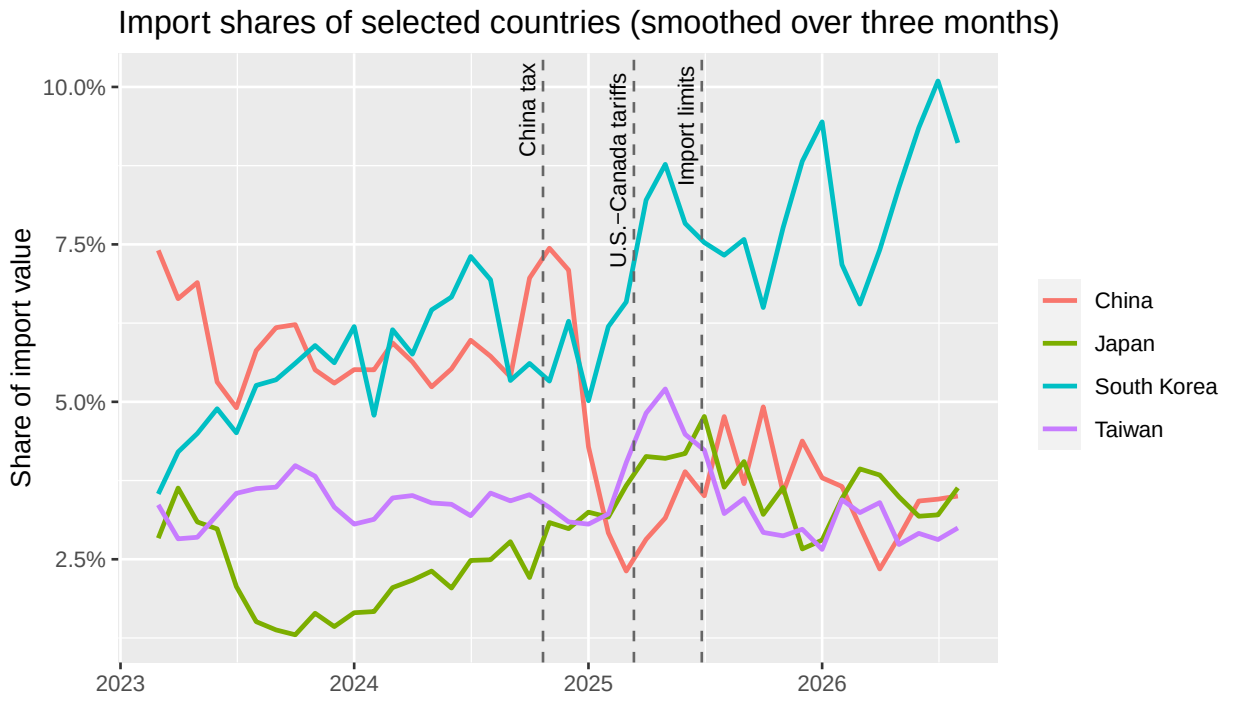
4. China's share over time



Before the tax, China's share moved between about 4% and 9%, drifting down in 2023 and sitting around 5% to 6% for most of 2024. It jumped to about 10% in September 2024, between the announcement and the start of the tax. Section 11 shows a similar jump for steel products that were not taxed, so the spike was probably not importers rushing to buy before the tax. After the tax started, China's share fell to about 2% in early 2025 and has mostly stayed between 2% and 4.5% since.

5. Which countries gained?

To make the trends easier to see, the chart below smooths each line by averaging over three months. The United States is left out because its share, around 50%, is on a very different scale.



Japan's and South Korea's shares were already growing before the tax: Japan's from about 1.3% in late 2023 to about 3% by October 2024, and South Korea's from about 3.5% in early 2023 to about 6% to 7% in 2024. So their later gains cannot all be credited to buyers switching away from China. Section 11 returns to this question with a sharper comparison.

6. Separating the tax from later policies

Because several policies changed after the tax, I split the period into five stages:

1. Before: January 2023 to July 2024
2. In between: August to October 2024, after the announcement but before the tax started
3. Tax only: November 2024 to February 2025
4. U.S.-Canada dispute: March to June 2025
5. Import limits: July 2025 to August 2026

The import limits did not cover every country at the same time. Limits on countries without a trade agreement with Canada, including China and Taiwan, began on June 27, 2025, so July 2025 is the first full month with limits. On August 1 they were extended to most trade-agreement partners, including Japan and South Korea, on different terms. The limits were changed again in December 2025 and June 2026, so stage 5 mixes several settings and should be read with care.

Table 3: Average share of Canada's steel imports (by value) in each stage

Country	1. Before	2. In between	3. Tax only	4. U.S.-Canada dispute	5. Import limits
China	5.9%	7.0%	3.7%	3.7%	3.6%
United States	52.0%	48.5%	53.2%	45.4%	49.3%

Country	1. Before	2. In between	3. Tax only	4. U.S.-Canada dispute	5. Import limits
South Korea	5.4%	5.6%	5.7%	8.0%	8.1%
Japan	2.2%	2.2%	3.6%	4.1%	3.5%
Vietnam	0.7%	0.3%	0.5%	0.7%	0.7%
Taiwan	3.4%	3.5%	3.2%	4.6%	3.1%
Other countries	30.3%	32.9%	30.1%	33.6%	31.7%

- **China's share fell as soon as the tax started**, from 5.9% to 3.7%, and was 3.7% and 3.6% in the two later stages. The later policies did not push it much lower.
- **Japan's share rose from 2.2% to 3.6%** during the tax-only stage, roughly matching China's drop. Because Japan's share was already rising, only part of this is likely to be buyers switching away from China.
- **The U.S. share fell later.** It was unchanged during the tax-only stage (53.2%) and dropped only during the U.S.-Canada dispute (45.4%), before recovering.
- **South Korea's and Taiwan's shares moved with the 2025 policies.** South Korea's share was flat during the tax-only stage and rose to 8.0% in the dispute stage. Taiwan's share went from 4.6% in the dispute stage to 3.1% once the import limits began. The limits reached Taiwan about a month before Japan and South Korea, and on different terms, which may help explain why these shares moved differently, but these averages cannot separate the policies.
- **Vietnam's share stayed small and steady.** Country totals like these cannot show whether Chinese steel reached Canada by way of other countries.
- **Suppliers outside these six kept a steady combined share**, about 30.3% before the tax and 30.1% in the tax-only stage, so China's lost share did not simply move to countries left out of this table.

7. Is it just the season?

The tax-only stage falls in winter, and steel buying can vary by season, so I compare the same winter months one year apart.

Table 4: Same winter months, one year apart (share by value)

country	Nov 2023 - Feb 2024 (before the tax)	Nov 2024 - Feb 2025 (with the tax)
China	5.8%	3.7%
Japan	1.7%	3.6%
South Korea	5.6%	5.7%
United States	52.6%	53.2%

In the same winter months, China's share fell from 5.8% to 3.7%, while South Korea's and the United States' shares barely changed. A purely seasonal explanation is therefore less likely, although a single year-on-year comparison cannot rule out a change in the seasonal pattern itself. Japan's share rose from 1.7% to 3.6%, but part of that continues its earlier rise.

8. Is it just prices?

Steel prices changed a lot over this period, so a falling value share could partly reflect cheaper Chinese steel rather than less of it. Here is the same comparison by weight:

Table 5: Same winter months, one year apart (share by weight)

country	Nov 2023 - Feb 2024 (before the tax)	Nov 2024 - Feb 2025 (with the tax)
China	5.1%	3.2%
Japan	0.7%	1.9%
South Korea	6.0%	6.3%
United States	51.4%	51.0%

The drop also shows up by weight: China's share fell from 5.1% to 3.2%, and Japan's rose from 0.7% to 1.9%. Japan's share by value is much higher than its share by weight, which suggests that steel from Japan costs more per tonne. Because a change in the mix of products can move value and weight shares differently, this check makes a pure price explanation less likely but cannot rule it out.

Imports themselves, not just shares

A share can fall because imports from China fell, because imports from everyone else rose, or both. This table compares the actual amounts in the same four winter months one year apart.

Table 6: Imports in Nov 2023 - Feb 2024 (before) and Nov 2024 - Feb 2025 (after)

Supplier	Value before (C$ million)	Value after (C$ million)	Change in value (%)	Weight before (1,000 tonnes)	Weight after (1,000 tonnes)	Change in weight (%)
All other countries	3544.7	3449.0	-2.7	2122.0	2061.0	-2.9
China	218.2	133.1	-39.0	115.1	69.6	-39.5

Imports from China fell by about 39% in value and 39% in weight compared with the same winter a year earlier, while imports from all other countries changed by -3% in value. So China's share fell mainly because Canada bought less from China, not just because it bought more from others. These are observed changes; demand and the mix of products also changed between the two winters.

9. Taking seasons and later policies into account together

Here I estimate how much China's share changed after the tax, after removing the usual seasonal pattern and allowing for the later U.S.-Canada dispute and import limits. The months between the announcement and the start of the tax are left out. The import-limit period starts in July 2025, the first full month after the limits on China began on June 27. The later extension to other countries in August is not separated.

The tables report a **p-value** for each estimate. It measures how easily ordinary month-to-month swings alone could produce a change as large as the one observed. By convention, a p-value below

0.05 is treated as unlikely to be chance. Because each month's figure is linked to the previous month's, I report two versions: the standard calculation, and one that allows for this link between nearby months.

Table 7: Change in China's share linked to each policy, after removing seasonal patterns (41 months)

Policy	Change in China's share (points)	Likely range: low	Likely range: high	p-value (standard)	p-value (allowing for linked months)
China tax	-2.413	-3.840	-0.986	0.009	0.002
U.S.-Canada dispute	0.194	-1.174	1.562	0.869	0.773
Import limits	-0.360	-1.181	0.460	0.685	0.375

After taking seasons and later policies into account, China's share is about 2.4 percentage points lower after the tax (for example, a fall from 6% to 3.6% is a drop of 2.4 points). In both versions of the calculation, the likely range for this drop excludes zero. The U.S.-Canada dispute and the import limits show no clear additional effect on China's share.

Allowing for a long-term drift

China's share was already drifting down in 2023, so I repeat the calculation allowing for a steady long-term drift.

Table 8: Effect of the tax with and without allowing for a long-term drift

Version	Change in China's share after the tax (points)	p-value (standard)	p-value (allowing for linked months)
Without long-term drift	-2.413	0.009	0.002
With long-term drift	-1.292	0.278	0.099

Once the drift is allowed for, the estimated drop shrinks to about 1.3 points and its likely range includes zero. With so few months after the tax, it is hard to tell a steady drift apart from a sudden drop. Allowing for the drift may also overcorrect, since China's share was roughly flat through 2024 and then fell sharply. This is why Section 11 uses a different approach: comparing taxed and untaxed products over the same months.

Checking the range of uncertainty by reshuffling the data

Another way to judge uncertainty is to reshuffle the data many times and see how much the estimate changes. A simple reshuffle treats each month as independent. A better version reshuffles blocks of

four consecutive months, so that months that belong together stay together.

Table 9: Likely range for the effect of the tax, calculated four ways

Method	Likely range: low (points)	Estimate (points)	Likely range: high (points)
Standard formula	-4.18	-2.41	-0.65
Formula allowing for linked months	-3.84	-2.41	-0.99
Simple reshuffle	-4.94	-2.41	0.11
Reshuffling blocks of 4 months	-4.54	-2.41	0.36

With the simple reshuffle, the likely range for the drop runs from -4.9 to 0.1 points; with blocks of four months, it runs from -4.5 to 0.4 points. Both ranges just include zero.

10. Two ways of averaging

China's share over a period can be measured in two ways: as the average of its monthly shares, which counts every month equally, or as China's total sales divided by total imports over the whole period, which gives more weight to busy months.

Table 10: China's share measured two ways

Period	Average of monthly shares	Total from China / total imports
Nov 2023 - Feb 2024	5.8%	5.8%
Nov 2024 - Feb 2025	3.7%	3.7%

Both methods give the same result (5.8% before and 3.7% after), so the finding does not depend on how the shares are averaged.

11. Comparing taxed and untaxed steel products

The tax did not cover every steel product. Of the 167 product types in the data, 130 were on the government's list and 37 were not. I checked each product against the official list; every listed item matches exactly one product type in the data.

The 37 untaxed product types are:

- 30 kinds of raw materials, such as pig iron, metal alloys used in steelmaking, scrap and iron powder;
- 7 kinds of finished steel, such as certain bars, angles and shapes.

If the tax caused China's share to fall, the drop should appear in taxed products but not in untaxed ones. Comparing the two groups month by month also cancels out anything that affected both groups at the same time, such as overall steel prices or Canadian demand.

Table 11: Taxed and untaxed products before the tax (Jan 2023 - Jul 2024)

Group	Product types	Imports from all countries (C$ billion)	Imports from China (C$ billion)	China's share
Not on list	37	1.630	0.077	0.047
On surtax list	130	18.415	1.113	0.060

Table 12: The seven untaxed finished-steel products, imports Jan 2023 - Jul 2024

Product code	Imports from all countries (C$ million)	Imports from China (C$ million)
7214.10	15.8	0.5
7216.61	38.1	3.4
7216.69	3.9	0.1
7216.91	21.6	7.4
7220.11	16.8	1.6
7228.10	11.6	0.9
7228.80	29.9	2.5

Before the tax, China supplied 6.0% of taxed products and 4.7% of untaxed products. The untaxed group is much smaller (about C$1.6 billion of imports over 19 months, compared with C$18.4 billion), so its monthly figures jump around more.

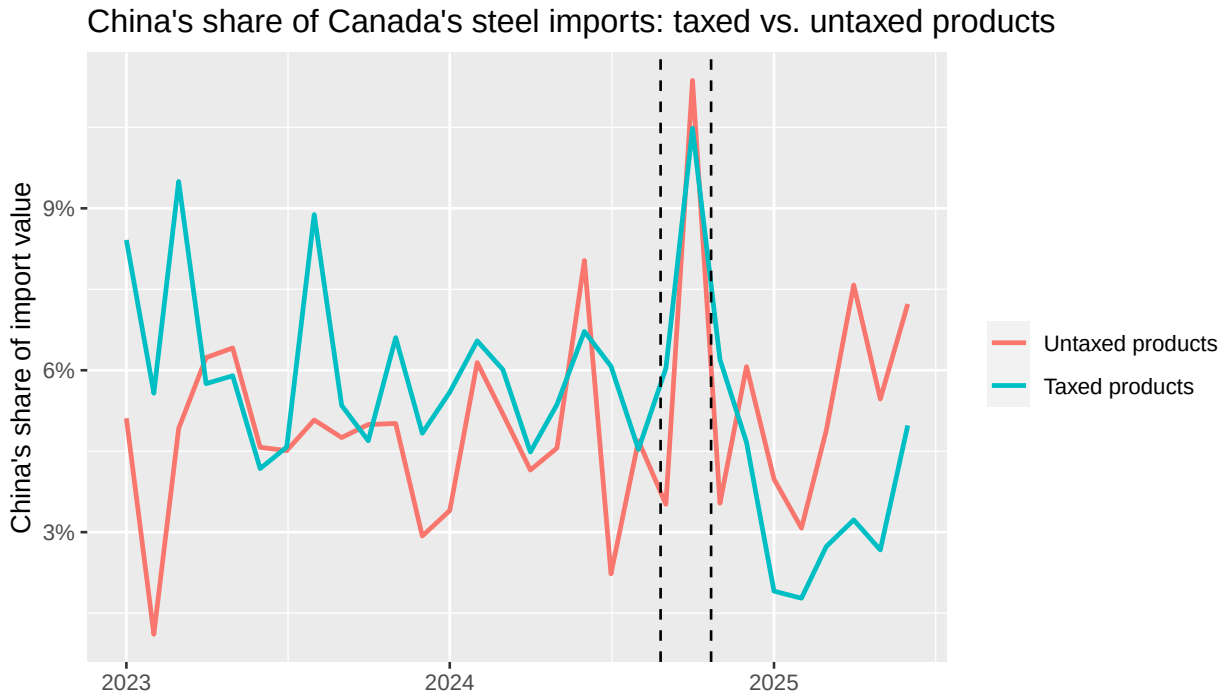


Table 13: How much more China's share fell for taxed products than for untaxed products

Months used	Extra drop for taxed products (points)	Likely range: low	Likely range: high	p-value (standard)
Jan 2023 - Jan 2025	-1.63	-4.13	0.88	0.19
Jan 2023 - Feb 2025 (main)	-1.88	-4.05	0.29	0.09
Jan 2023 - Jun 2025	-3.06	-4.71	-1.42	0.00

Before August 2024, China's shares in the two groups moved roughly together, between about 4% and 7%. Both jumped in September 2024, so that jump was not specific to taxed products. After the tax started, China's share of taxed products fell to about 2% to 3% and stayed there, while its share of untaxed products returned to around 4% to 7%.

The main comparison uses only the four months after the tax and before the U.S.-Canada dispute. It finds that China's share of taxed products fell by about 1.9 percentage points more than its share of untaxed products. Using months up to June 2025 gives a larger drop of about 3.1 points, but that period includes the dispute, when Canada also put tariffs on U.S. steel, so other changes could be mixed in.

February 2025 may already have been affected, because the U.S. announced and then paused tariffs on Canadian goods in early February and announced steel tariffs on February 10. Ending the comparison in January 2025 gives a similar drop of about 1.6 points, so February does not drive the result.

Allowing for links between months

The p-values in the table above treat each month as unrelated to the next. In reality, China's share in one month is linked to the month before, which can make the standard calculation misleading. Here I repeat the comparison with a calculation that allows for this. The estimated drops stay the same; only the measure of uncertainty changes.

Table 14: The same comparison, allowing for links between months

Months used	Extra drop for taxed products (points)	Likely range: low	Likely range: high	p-value (standard)	p-value (allowing for linked months)
Jan 2023 - Jan 2025	-1.626	-3.557	0.304	0.191	0.094
Jan 2023 - Feb 2025 (main)	-1.882	-3.604	-0.161	0.086	0.034
Jan 2023 - Jun 2025	-3.061	-4.826	-1.297	0.001	0.001

With this calculation, the main comparison's p-value falls below 0.05, and the comparison through June 2025 is even clearer. This should not be over-read. With only about two years of monthly data, this calculation is itself approximate and can be too optimistic, and the reshuffling ranges in Section 9 still include zero. So far, the main result sits right at the edge: depending on the method, it falls just on one side or the other of the usual 0.05 line. The next two parts check how much this depends on choices in the calculation.

How much does the result depend on calculation choices?

The calculation that allows for links between months needs a choice: how many neighbouring months are allowed to be linked. I used three. The first table repeats the calculation with other choices. The second leaves out each of the four months after the tax in turn.

Table 15: Main comparison with different numbers of linked months

Neighbouring months linked	Likely range: low (points)	Likely range: high (points)	p-value
0	-4.005	0.240	0.079
1	-3.912	0.147	0.067
2	-3.773	0.009	0.051
3	-3.604	-0.161	0.034
4	-3.481	-0.284	0.023
6	-3.317	-0.448	0.013

Table 16: Main comparison leaving out one month after the tax

Month left out	Extra drop for taxed products (points)
November 2024	-2.95
December 2024	-1.59
January 2025	-1.37
February 2025	-1.63

The answer depends on the choice. With three or more linked months the p-value is below 0.05, but with two or fewer it is not (p-values from 0.013 to 0.079 across the choices). Leaving out one month at a time, the estimated drop ranges from about 1.4 to 2.9 points. Its direction does not depend on any single month, but its size does: leaving out November 2024, when the gap had not yet widened, gives the largest drop.

Comparing product by product

The comparisons so far add up all taxed products into one group and all untaxed products into another, which leaves only two lines to compare. Another approach keeps each of the 167 products separate. It compares China's share in each product before and after the tax, allows every product to have its own usual level, and judges uncertainty by how much the results vary from product to product rather than from month to month. This allows for links between months within each

product, but not for shocks that hit all products in the same month, so it is a different check rather than a strictly better one. Larger products count for more, in proportion to their imports before the tax, so that the result is comparable with the group totals above.

Table 17: Product-by-product comparison (167 products, uncertainty measured across products)

Months used	Larger products count more	Extra drop for taxed products (points)	Likely range: low	Likely range: high	p-value (across products)
Jan 2023 - Jan 2025	Yes	-1.783	-4.238	0.672	0.155
Jan 2023 - Feb 2025 (main)	Yes	-1.612	-3.835	0.611	0.155
Jan 2023 - Jun 2025	Yes	-2.642	-4.457	-0.828	0.004
Jan 2023 - Jan 2025	No	0.519	-2.745	3.783	0.755
Jan 2023 - Feb 2025 (main)	No	0.736	-1.988	3.461	0.596
Jan 2023 - Jun 2025	No	-2.093	-4.444	0.257	0.081

The product-by-product comparison points the same way as the group comparison: in the main period, China's share of taxed products fell by about 1.6 points more than its share of untaxed products. But once uncertainty is judged across products, its likely range includes zero ($p = 0.155$); only the longer comparison through June 2025, which overlaps the U.S. dispute, stands out ($p = 0.004$). Research on this kind of comparison recommends measuring uncertainty across the units that were or were not affected (Bertrand, Duflo and Mullainathan, 2004), so this result is a strong reason not to call the drop clear on four months of data.

When every product counts equally, whatever its size, the drop disappears in the main period (+0.7 points). Most of the 167 products are small, and China's share of a small product can swing widely from month to month, so this version is very noisy. The two versions answer different questions: the weighted one describes steel imports as a whole, while the unweighted one describes the typical product. The gap between them suggests that the drop came mainly from the large products, although this check alone cannot prove it.

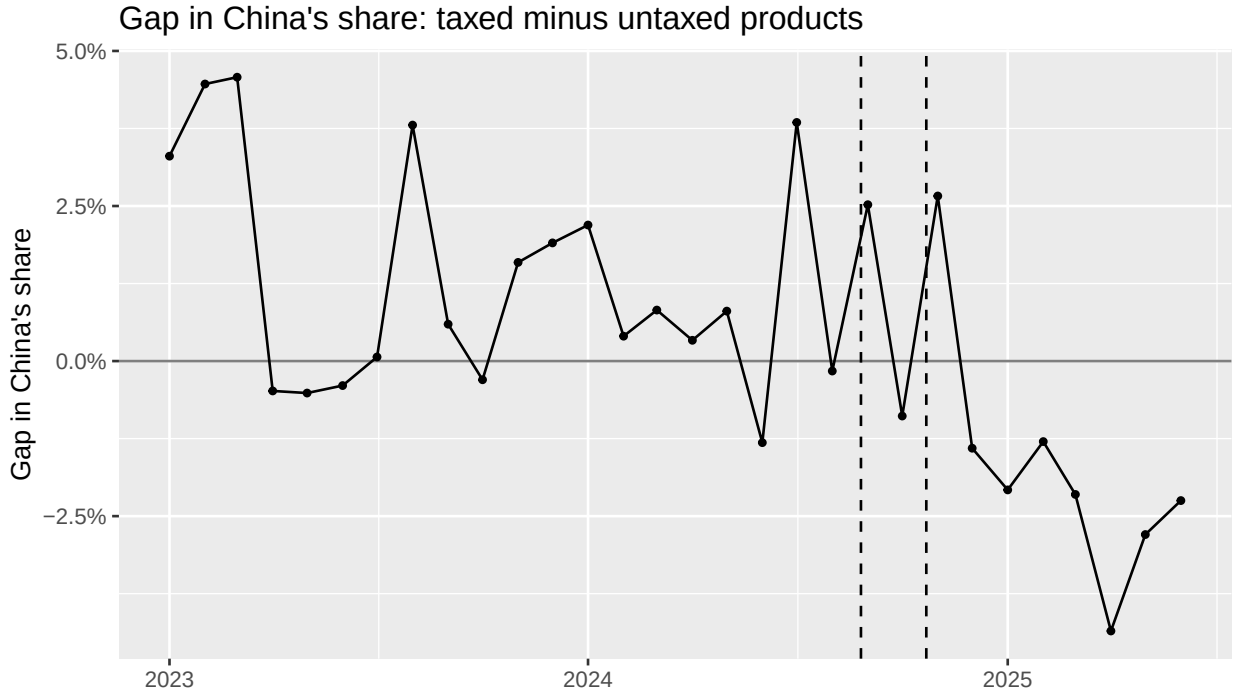
Did the two groups move together before the tax?

This comparison only works if China's shares in taxed and untaxed products would have moved together without the tax. That cannot be proven, but two checks on the period before the tax (January 2023 to July 2024) can reveal problems:

1. **Was the gap already changing?** If the gap between the groups was steadily widening or narrowing before the tax, that would be a warning sign.
2. **A fake start date.** If I pretend the tax started in January 2024 and use only data from before the real tax, I should find no effect.

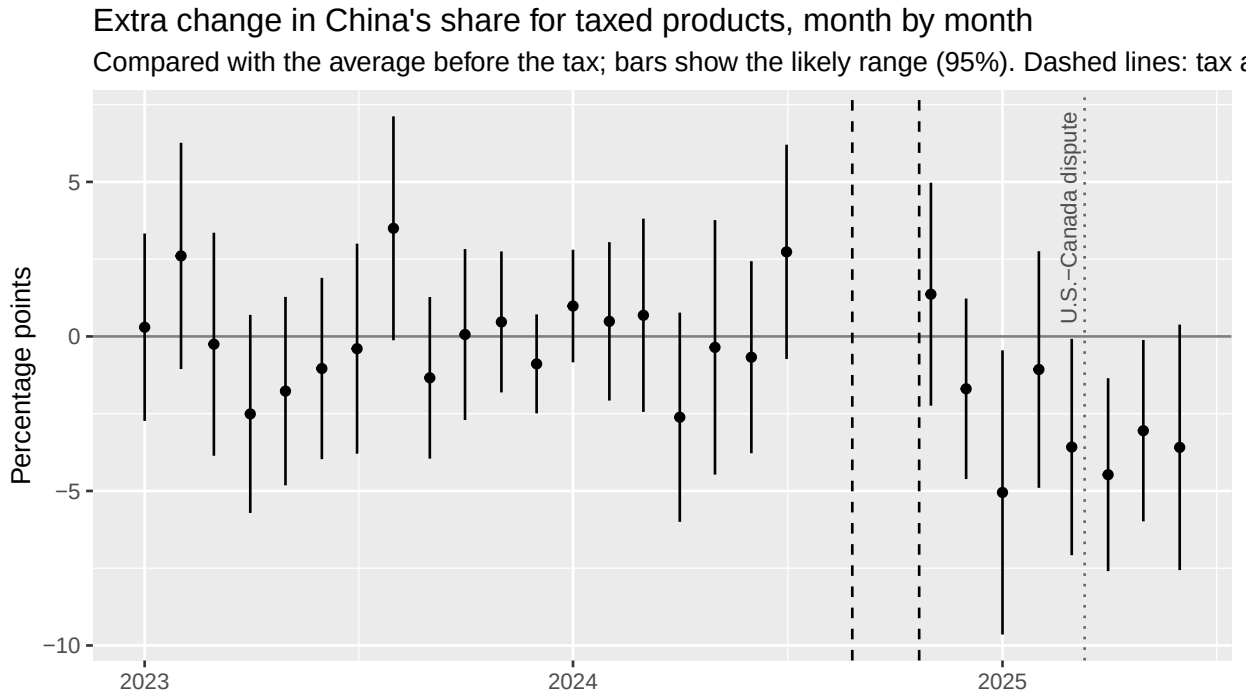
Table 18: Checks on the period before the tax (Jan 2023 - Jul 2024)

Check	Size (points)	p-value
Change in the gap per month	-0.087	0.304
Fake tax start in January 2024	-0.539	0.459



Before the tax, the gap between the two groups changed by only about 0.09 points per month, and the fake start date shows a change of only about 0.5 points. Neither is large relative to its uncertainty, so neither check finds a warning sign. Two cautions apply: with only 19 months, these checks can catch only fairly large differences, and if the small downward drift in the gap were real and continued, it would explain part of the measured drop.

The product-by-product data allow a month-by-month version of the same check. For each month, it estimates how far China's share of taxed products was above or below its share of untaxed products, compared with the average gap before the tax. Before the tax, the points should hover around zero.



Before the tax, the monthly points move up and down around zero without a clear pattern, and taken together they are consistent with a steady gap ($p = 0.290$). After the tax, the drop did not appear at once: in November 2024 the gap was still close to its usual level, and it widened in December and especially January 2025 before narrowing again in February. One possible reason is that steel ordered before the tax was still arriving in the first weeks, although the data do not show order or delivery dates. From March 2025 the points stay below zero, but those months also include the U.S.-Canada dispute.

These checks have a limit. If the calculation also allows the gap to follow a steady straight-line trend, the drop in the main period shrinks to about 0.5 points ($p = 0.635$; China's row in the last table of this section). The trend is estimated from all the months used, before and after the tax, not only from the months before it. Even so, with only four clean months after the tax, the data cannot tell a sudden drop apart from a slow drift that simply continued.

A closer comparison group: untaxed finished steel

Most untaxed products are raw materials, which may be bought for different reasons than the finished steel on the tax list. As a check, I repeat the comparison using only the seven untaxed finished-steel products. They are more similar to the taxed products, but the group is much smaller, so its figures jump around more.

Table 19: Taxed products and untaxed finished steel before the tax

Group	Imports from all countries (C$ million)	China's share
Not on list	137.639	0.118
On surtax list	18414.597	0.060

Table 20: Comparison with untaxed finished steel only

Months used	Extra drop for taxed products (points)	Likely range: low	Likely range: high	p-value (allowing for linked months)
Jan 2023 - Jan 2025	-4.177	-8.489	0.134	0.057
Jan 2023 - Feb 2025 (main)	-4.563	-8.462	-0.664	0.024
Jan 2023 - Jun 2025	-6.817	-10.938	-2.696	0.002

With untaxed finished steel as the comparison, China's share of taxed products fell by about 4.6 points more in the main period and about 6.8 points more through June 2025. The drop is in the same direction as before but larger. This group is very small (about C$140 million of imports before the tax), and China already supplied twice as large a share of it. Chinese exporters may also have shifted some sales toward these similar but untaxed products, which would make the measured effect too large. So I treat this as a check rather than the main result.

Which countries gained in the taxed products?

Section 5 could not tell which countries took China's place, because Japan's and South Korea's shares were already rising for reasons unrelated to the tax. The same comparison of taxed and untaxed products can be applied to every supplier. Because the shares of all suppliers add up to 100% in each group, the changes across suppliers add up to zero, so China's loss must show up as someone's gain. "Other countries" covers everyone outside the six suppliers downloaded separately, about 30% of Canada's steel imports. As a check, a second version also allows each supplier's gap to follow its own steady trend, estimated over all the months used. Because seven suppliers are tested at once, one of them could look clear by luck, so the table also adjusts the p-values for the number of suppliers tested (Holm's method).

Table 21: Change in each supplier's share of taxed vs. untaxed products, Jan 2023 - Feb 2025

Supplier	Share of taxed products before the tax (%)	Extra change for taxed products (points)	p-value	p-value adjusted for 7 suppliers	With each supplier's own trend (points)	Adjusted p-value (with trend)
China	6.04	-1.88	0.034	0.202	-0.48	1.000
United States	52.11	-3.40	0.338	1.000	2.77	1.000
South Korea	5.73	1.54	0.111	0.445	-1.61	0.440
Japan	2.38	1.56	0.003	0.019	2.09	0.073
Vietnam	0.76	-0.19	0.042	0.208	0.08	1.000
Taiwan	3.68	-0.01	0.971	1.000	-0.08	1.000

Supplier	Share of taxed products before the tax (%)	Extra change for taxed products (points)	p-value	p-value adjusted for 7 suppliers	With each supplier's own trend (points)	Adjusted p-value (with trend)
Other countries	29.29	2.38	0.511	1.000	-2.77	1.000

In taxed products, where China's share fell, the clearest gain went to **Japan**: its share of taxed products rose by about 1.6 points more than its share of untaxed products. This stays clear after adjusting for testing seven suppliers (adjusted p = 0.019). With its own trend added, the gain is still about 2.1 points, but after the adjustment it is no longer clear (adjusted p = 0.073). Japan sold almost no untaxed steel products to Canada, so this mainly reflects its growing share of taxed products. South Korea's share of taxed products also rose, by about 1.5 points, but the estimate is not precise (p = 0.111), and it disappears once Korea's earlier trend is allowed for. The estimates for the United States and other countries are large in either direction and far from clear, so the data cannot say whether they gained. Vietnam's and Taiwan's shares changed by less than a quarter of a point.

So Japan is the most likely gainer, but even its result is not conclusive, and it does not show that the buyers who stopped buying from China switched to Japan. Most of the uncertainty sits with the United States and other countries, which together supplied about 81% of taxed products before the tax.

12. Limitations

- **Very little clean data.** Only four months separate the tax from the U.S.-Canada dispute, so the cleanest estimates are uncertain. February 2025 may already reflect worries about U.S. tariffs.
- **Long-term drift.** Looking at China's share on its own, the estimated drop shrinks and its likely range includes zero once a steady long-term drift is allowed for. Comparing taxed and untaxed products avoids this problem only if the two groups would otherwise have moved together. The checks found no sign that they would not, but they can catch only fairly large differences, and allowing the gap to follow a steady straight-line trend shrinks the main estimate to about 0.5 points.
- **Comparison group.** Most untaxed products are raw materials that serve different buyers from finished steel. Using only untaxed finished steel gives a larger drop in the same direction, but that group is very small and may have picked up some Chinese sales diverted from taxed products.
- **Uncertainty.** Each month's figure is linked to the previous month's. I used calculations that allow for this, but with so few months after the tax, all measures of uncertainty are approximate. The result also depends on how many neighbouring months are allowed to be linked, and in the comparison that keeps products separate the likely range of the main result includes zero.
- **Size of products.** The drop appears when larger products count for more, as they do in China's overall share, but not when every product counts equally.

- **Coverage.** The tax also applies to some steel products not studied here, such as pipes and tubes.
- **Country of origin.** Customs records show the country of origin. Simple transshipment does not normally change origin. Processing in a third country may change recorded origin depending on the applicable rules; these data cannot identify such routes. Canada's July 2025 rule on steel first melted and poured in China suggests this was a concern.
- **What the results do not show.** A lower Chinese share does not mean the tax was good for Canada overall. I did not look at Canadian steel production, jobs, costs for businesses that use steel, or prices paid by consumers.
- **Complete records, not a sample.** Customs data cover every import rather than a survey sample, so the uncertainty comes from month-to-month swings, not from sampling.

13. Related research

Other studies of tariffs point in the same direction as these results:

- **Buyers switch to other suppliers.** After the United States put duties on washing machines from South Korea and later China, production moved to other exporting countries (Flaaen, Hortaçsu and Tintelnot, 2020). During the 2018-2019 U.S.-China trade war, many countries that were not part of the dispute increased their exports of the targeted products (Fajgelbaum and others, 2024).
- **The taxed country sells elsewhere.** A U.S. duty on Japanese products raised Japan's exports of those products to other countries (Bown and Crowley, 2007). This supports the caution in Section 11 that Chinese exporters may have shifted sales toward similar untaxed products.
- **A falling import share is not the same as a gain.** Studies of the 2018 U.S. tariffs found that buyers in the United States bore most of the cost through higher prices (Amiti, Redding and Weinstein, 2019; Fajgelbaum and others, 2020). This is why a smaller Chinese share should not be read as proof that Canada gained overall.
- **Why I was careful about certainty.** Comparisons like the one in Section 11 can overstate how certain the results are when monthly figures are linked over time (Bertrand, Duflo and Mullainathan, 2004), which is why I also used calculations that allow for these links. Checks of whether two groups moved together before a policy often miss real differences (Roth, 2022), which is why I treat those checks as reassuring but not decisive.

14. Conclusion

The results fit the basic economic idea that making one supplier more expensive pushes buyers toward others. China's share of Canada's steel imports fell sharply when the tax started. The drop also shows up by weight, remains when comparing the same winter months one year apart, and happened mainly in the products the tax covered. The main difference-in-differences estimate is a 1.9-percentage-point decline, roughly a third of China's earlier share. This is consistent with a surtax effect, but sensitive to modelling choices. Comparing taxed and untaxed products, the estimates range from about 1.6 to 3.1 points depending on the months and method used. A narrower comparison group gives a larger drop, and allowing for a steady trend in the gap gives a much smaller one. The evidence is suggestive rather than conclusive: with only four clean months after the tax, the main estimate is imprecise, and under some calculations, including the product-by-product comparison, its likely range includes zero.

Where China's lost share went is only partly clear. Comparing taxed and untaxed products, Japan is the most likely gainer, though not conclusively; South Korea's estimated gain is imprecise, and the United States and smaller suppliers are too noisy to tell. Later changes in the U.S., South Korean and Taiwanese shares line up with the 2025 trade dispute and import limits. Separating the tax from these overlapping trends and policies was the hardest part of this analysis.

Technical notes

For readers familiar with the methods:

- **Main design:** difference-in-differences comparing 130 listed and 37 unlisted six-digit HS products (chapter 72), with month fixed effects. Main window (January 2023 to February 2025): -1.9 percentage points; p = 0.086 with OLS standard errors and p = 0.034 with Newey-West standard errors (3 lags). Window ending January 2025: -1.6 points (Newey-West p = 0.094). Window ending June 2025: -3.1 points (Newey-West p = 0.001).
- **Sensitivity of the main window:** Newey-West p-values with 0, 1, 2, 3, 4 and 6 lags: 0.079, 0.067, 0.051, 0.034, 0.023, 0.013. Leaving out one post-tax month at a time: -2.9 to -1.4 points.
- **Absolute imports**, November-February one year apart: imports from China -39.0% in value and -39.5% in weight; all other countries -2.7% in value.
- **Pre-trend checks (January 2023 to July 2024):** linear trend in the listed-minus-unlisted gap of -0.09 points per month (p = 0.30); placebo treatment in January 2024 of -0.5 points (p = 0.46).
- **Product-level panel:** 167 six-digit products, product and month fixed effects, weighted by average monthly imports over the 19 pre-tax months, standard errors clustered by product. Main window: -1.6 points (p = 0.155); to January 2025: -1.8 points (p = 0.155); to June 2025: -2.6 points (p = 0.004). Unweighted, main window: +0.7 points (p = 0.596).
- **Event study** on the same panel, months relative to the pre-tax average: joint test that all 19 pre-tax months are equal, chi-squared = 20.8 with 18 degrees of freedom (p = 0.290); average November 2024 to February 2025: -1.6 points.
- **Gap-specific linear trend**, fitted jointly over pre- and post-tax months (listed minus unlisted, main window, Newey-West): -0.5 points (p = 0.635).
- **Diversion** (same comparison for each supplier's share, main window, Newey-West; the estimates sum to zero by construction): Japan +1.6 points (p = 0.003, Holm-adjusted for seven suppliers p = 0.019; with its own trend +2.1, p = 0.010, Holm-adjusted p = 0.073); South Korea +1.5 points (p = 0.111); United States -3.4 points (p = 0.338); other countries +2.4 points (p = 0.511).
- **Alternative control group** (seven unlisted finished-steel products): -4.6 points in the main window (Newey-West p = 0.024).
- **China-only time series** with calendar-month fixed effects and cumulative policy indicators: surtax coefficient -2.4 points (OLS p = 0.0096; Newey-West p = 0.002); with a linear trend -1.3 points (Newey-West p = 0.099). Moving block bootstrap (block length 4) 95% interval: -4.5 to +0.4 points. The import-limit indicator starts in July 2025.
- **Other checks:** same-season comparison, tonnage shares, and a ratio estimator compared with the mean of monthly shares. The input files are checked for missing months, missing or negative values, unexpected units and duplicate rows.

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